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Title of Project : Fit-Track AI
Real-Time Exercise calculation & Burnt Calories Prediction System

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Problem statement:

- Many people perform exercises at home without tracking their actual performance.
- Uncounted or incorrect repetitions lead to ineffective workouts and poor results.
- Most fitness applications provide only general calorie estimates and ignore exercise-specific data.
- There is a need for a system that can **count exercise repetitions and predict calories burnt based on the performed exercise in real-time**.

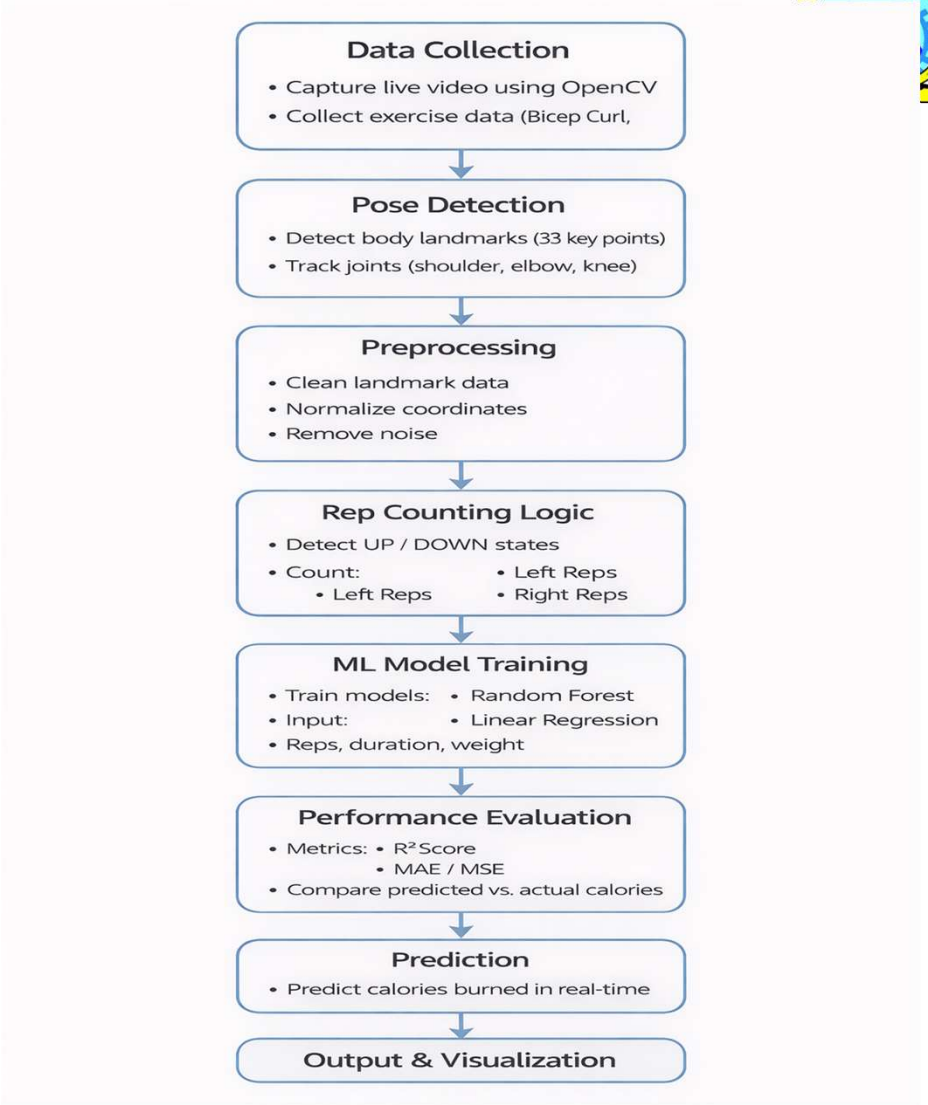


How we are solving the problem?

- We propose a **Fit-Track** using computer vision and machine learning.
- The system uses a **webcam** to capture user exercises in real time.
- It detects body joints using **pose estimation (MediaPipe / BlazePose).**
- Identifies key **body landmarks** such as shoulders, elbows, hips, knees, and ankles.
- Calculates **joint movements** to track exercise performance accurately.
- Automatically **counts exercise repetitions** based on motion patterns.
- **Predicts calories burnt** based on the performed exercise and reps.
- Provides **real-time feedback** to improve workout effectiveness.

Project Phases:

Phase 1: Data Collection
Phase 2: Preprocessing
Phase 3: Feature Extraction
(Rep counting Logic)
Phase 4: ML Model Training
Phase 5: Performance Evaluation
Phase 6: Prediction



Rep Counting Architecture



- **Input Processing (OpenCV – Video Capture):**

Captures live video of the user performing exercises through a webcam.

- **Pose Detection (Mediapipe/Blazepose Model)[1]:**

Detects 33 body landmarks such as shoulders, elbows, hips, knees, and ankles.

- **Feature Extraction :**

Calculates joint angles using the coordinates of detected body landmarks (e.g., elbow angle for bicep curl, knee angle for squat).

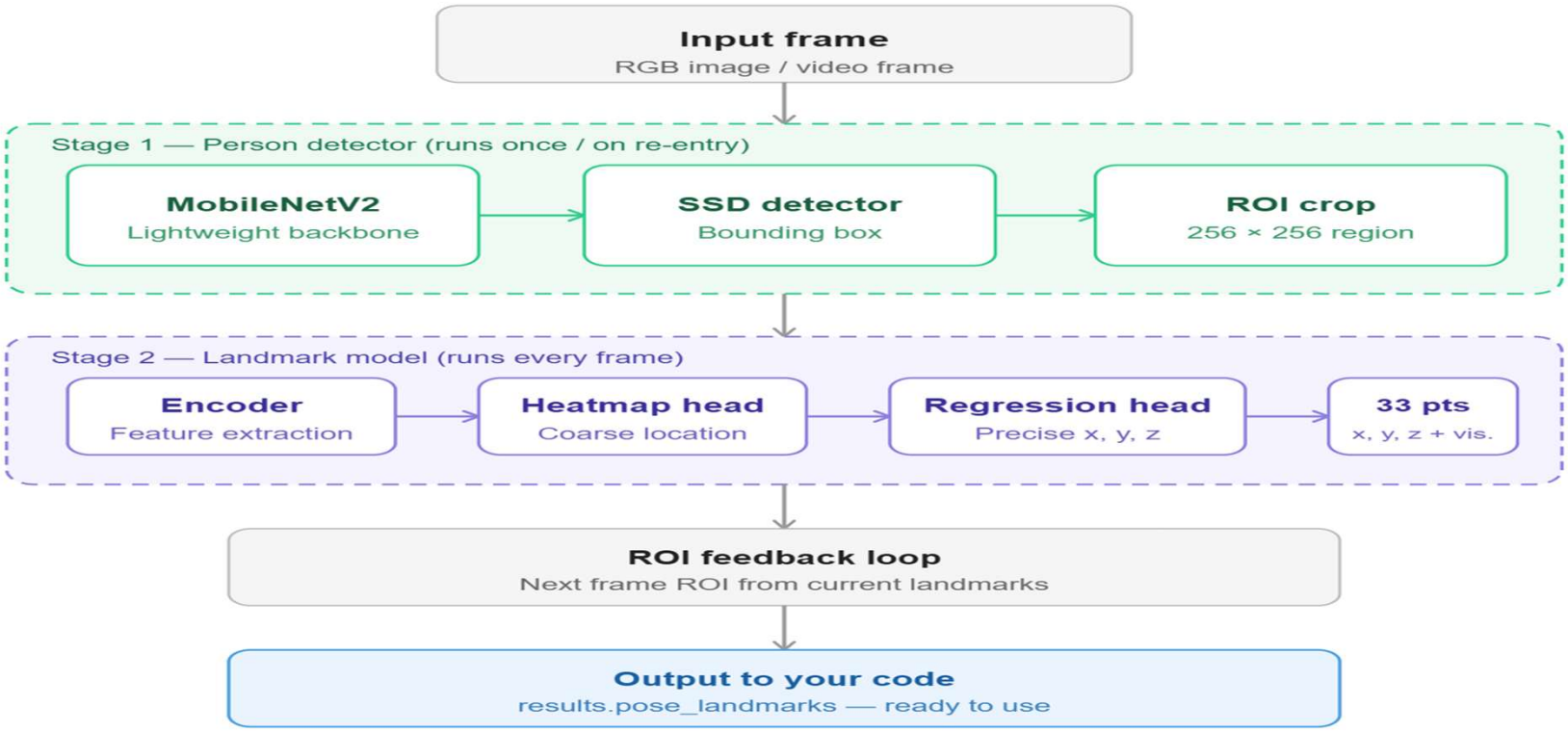
- **Rep Counting Logic:**

Analyzes angle changes to detect movement stages (**UP and DOWN positions**).
A repetition is counted when a full movement cycle (DOWN → UP → DOWN) is completed.

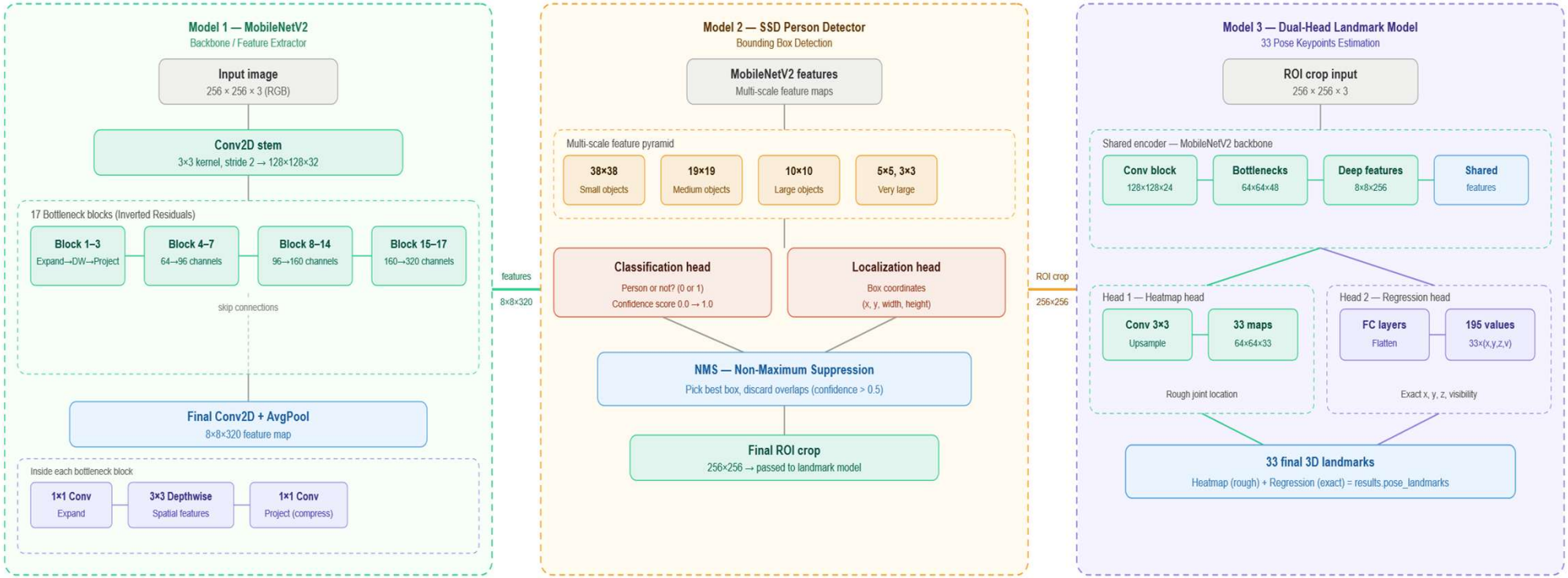
- **Output (Rep Counter):**

Displays the number of repetitions performed for each exercise (Left reps, Right reps, Squats).

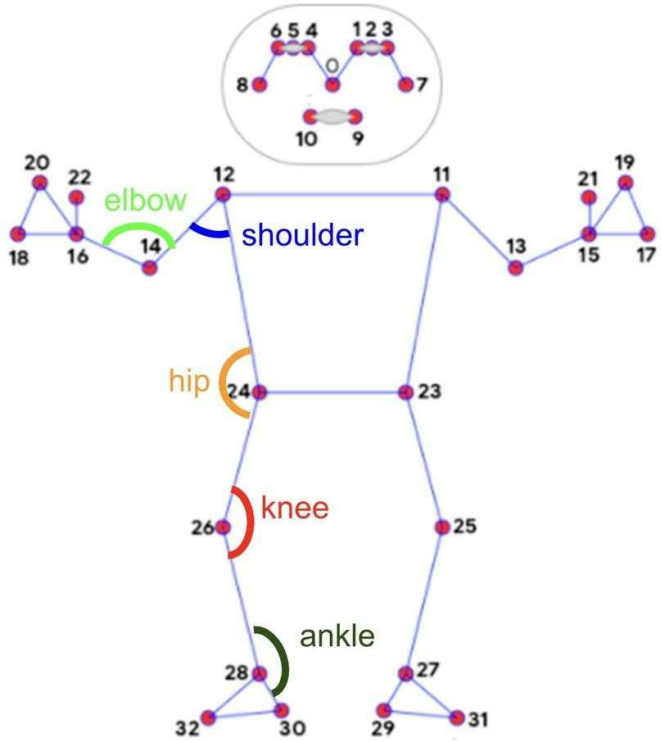
Mediapipe Architecture:(🔗 source)



Mediapipe(BlazePose) PipeLining:



Body Landmarks and Joint Angle Calculation:



Rep Counter Mechanism:

- **Pose Detection (MediaPipe):** Detects 33 body landmarks and provides (x, y) coordinates of joints
- **Angle Calculation:**

  $$\theta = |\text{atan2}(y_3 - y_2, x_3 - x_2) - \text{atan2}(y_1 - y_2, x_1 - x_2)| \times 180/\pi$$

- **Rep Counting Logic:**

  If angle > 150° → DOWN (arm straight)

  If angle < 60° → UP (arm bent)

- **Repetition Rule:**

  DOWN → UP → DOWN = 1 rep

- **Output:** Displays total repetitions in real-time

Burnt Calories Prediction:

- **Input (Exercise Data):** Takes reps, exercise type, and duration from rep counter
- **User Details** Includes weight, age, height, and gender[1]
- **Feature Preparation:** Combines all inputs into a structured format
- **Models Used:[2](Model Comparison)**

  1. Random Forest
  2. Linear Regression
  3. XGBoost
  4. SVM

- **Prediction:** Model predicts calories burned based on input data
- **Output:** Displays estimated calories burned

Datasets:

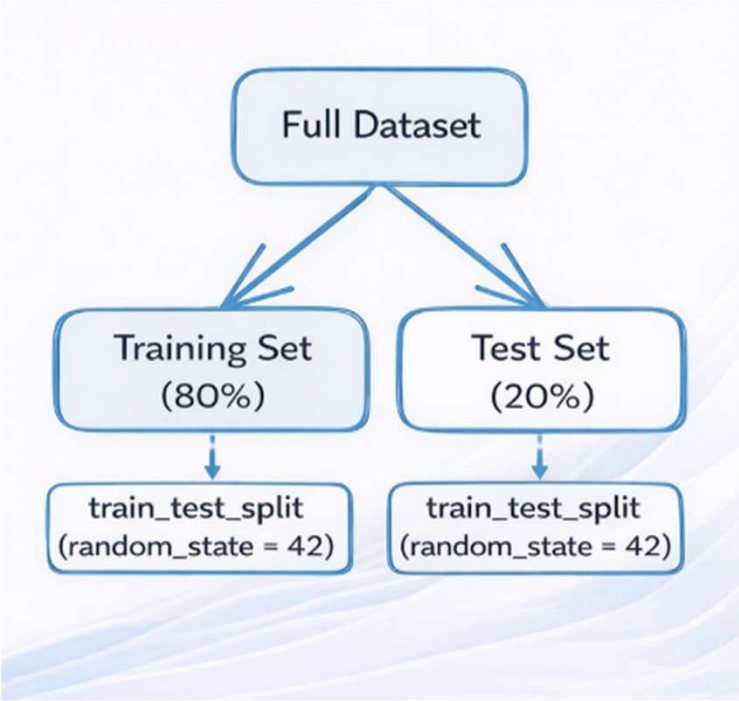
- **Dataset Name:** Exercise & Calories Burn Datasets [3]
- **Total Samples:** 16000 records
- **Number of Features:** 8
  (exercise_type, gender, age, intensity, body_weight_kg, duration_secs, reps, calories_burned)
- **Input Features:** exercse_type, intensity, body_weight_kg, duration_secs, reps
- **Target Variable:** calories_burned

Dataset Splitting Strategy:

➤ The dataset was split into two mutually exclusive subsets:

Training set: 80%

Test set: 20%



➤ **Key Properties of the Split:**

- Dataset split using **train_test_split**
- Training set used for **model learning**
- Test set used for **final evaluation**

Linear Regression Alogorithms:

Model: $\hat{y} = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$

Weights: $W = (X^TX)^{-1}X^Ty$

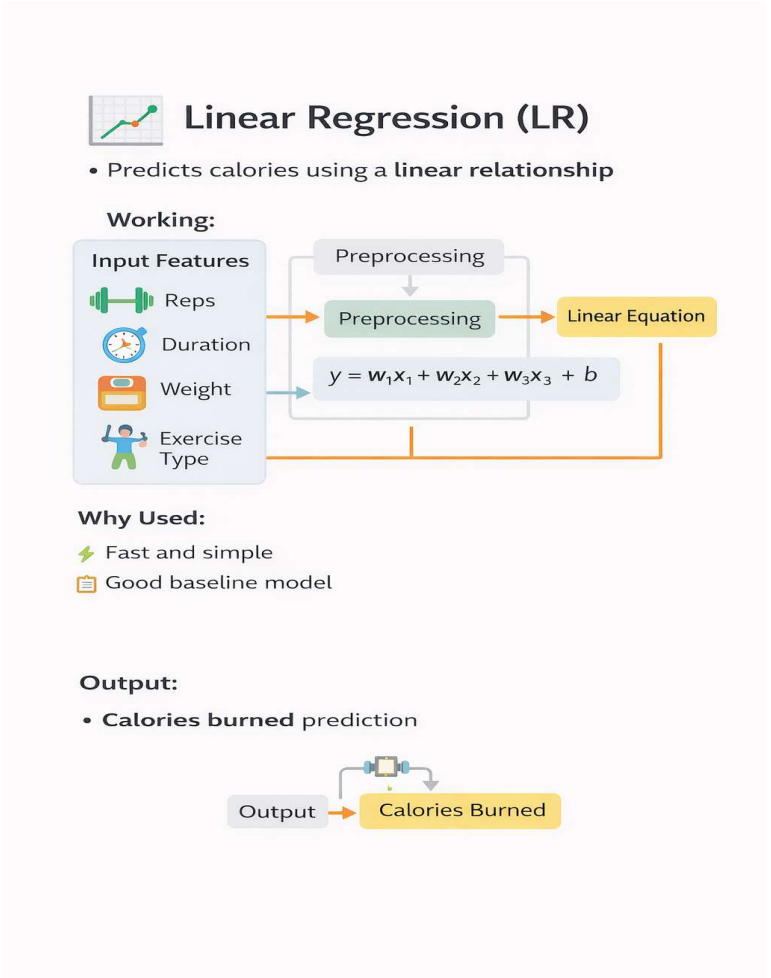
$$w_i = \frac{\partial \hat{y}}{\partial x_i}$$

Objective: $\min \sum (y - \hat{y})^2$

Idea: Linear mapping of features → Calories prediction

Output:

$\hat{y} = Calories\ Burned$



Random Forest Algorithms:

Input Features:

$X = [x_1, x_2, x_3, x_4] = [Reps, Duration, Weight, Exercise\ Type]$

Tree Predictions: $T_i(X) = \hat{y}_i, i = 1,2,\ldots,N$

Final Prediction (Core Formula):

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i(X)$$

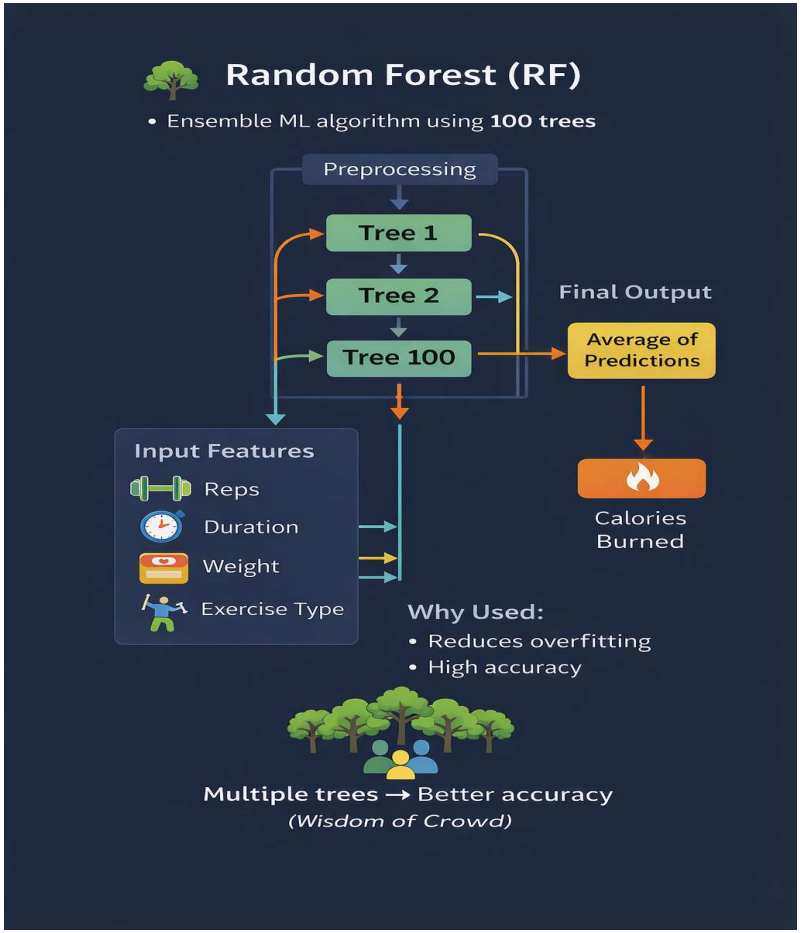
For 100 Trees: $\hat{y} = \frac{1}{100}\sum_{i=1}^{100} \hat{y}_i$

Variance Reduction: $Var(\hat{y}) = \frac{1}{N}\sum_{i=1}^{N} Var(T_i)$

Output: $\hat{y} = Calories\ Burned$

Key Idea: Prediction $=$ *Average of Multiple Trees*

(Ensemble Learning)



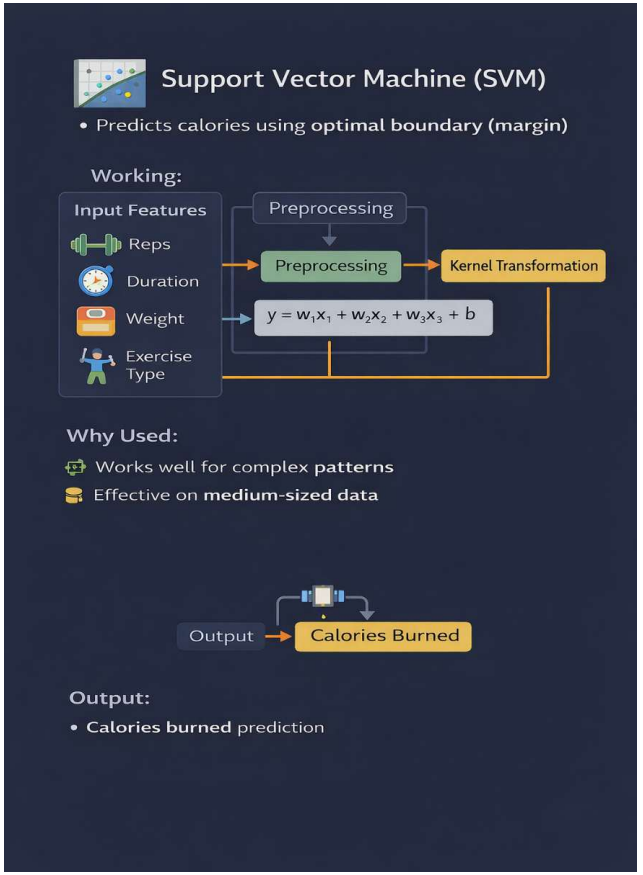
Support Vector Machine (SVM):

Idea:
Predicts calories using an **optimal hyperplane (boundary)**
Model: $w \cdot x + b = 0$

With Kernel Trick: $\hat{y} = \sum_{i=1}^{n} \alpha_i \, K(x_i, x) + b$
Objective: $\min \frac{1}{2} \, || \, w \, ||^2$
(Maximizes margin between data points)

Working: Input → Kernel Transform → Optimal Boundary → Prediction
Why Used:
- Handles complex (non-linear) data
- Effective for medium datasets

Output:
$\hat{y} = Calories\ Burned$



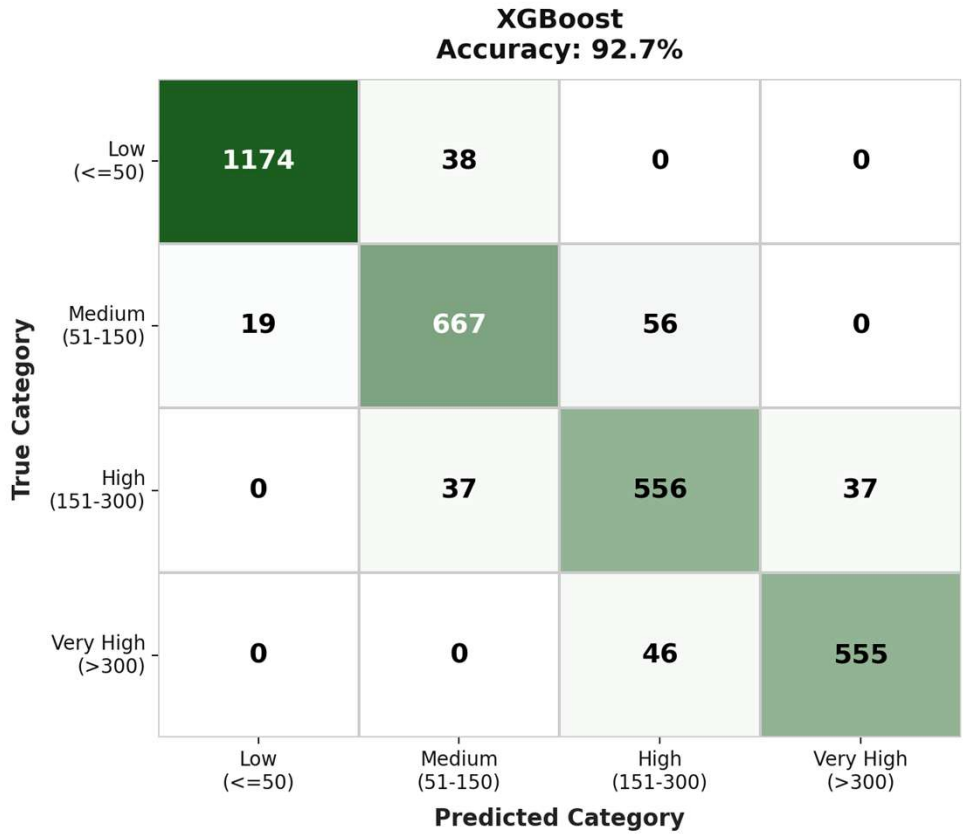
XGBoost (Extreme Gradient Boosting)

Idea:
Predicts calories using **sequential boosted trees**
Model: $F(x) = \sum_{m=1}^{M} \gamma_m \, h_m(x)$

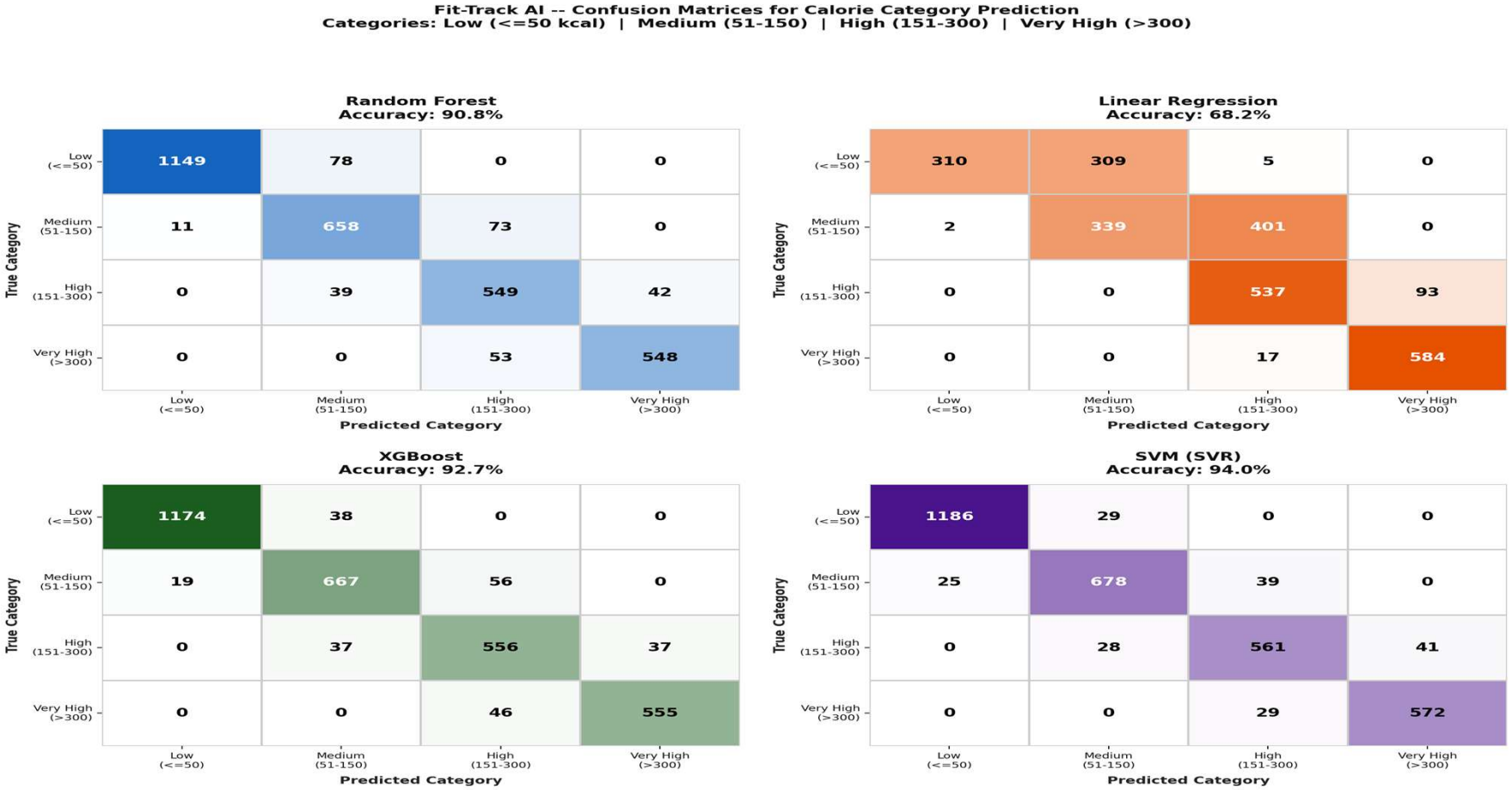
Update Rule: $F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$
Working: Input → Sequential Trees → Error Correction → Final Prediction
Loss Minimization: $\min \sum \left(y - F(x)\right)^2$
Why Used:
- High accuracy
- Learns from previous errors

Output:

$\hat{y} = Calories\ Burned$



Confusion Matrices:

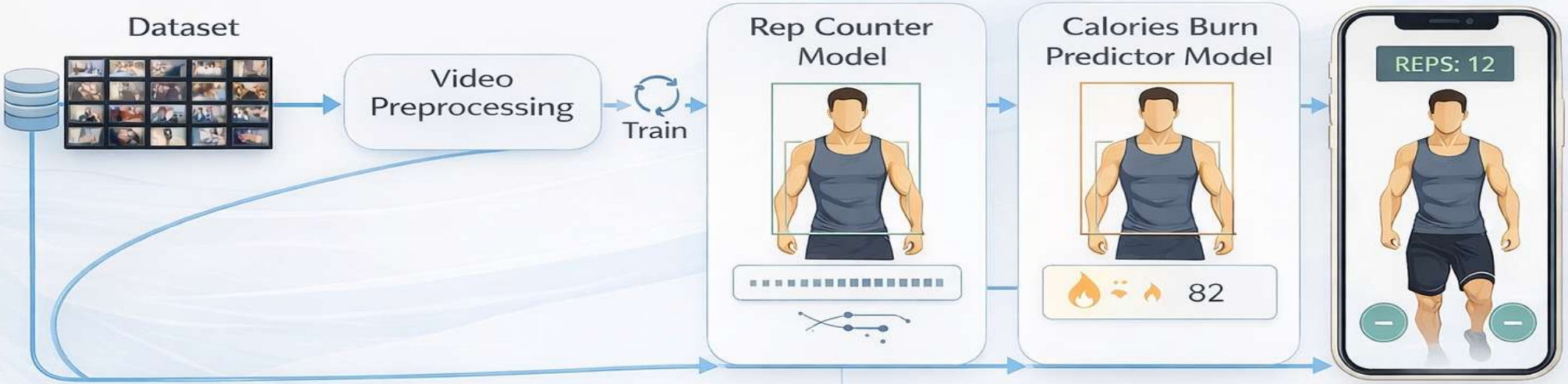


Models Accuracy:

Fit-Track AI -- Model Accuracy Metrics Summary Table

Model	Train R2 (%)	Test R2 (%)	MAE (kcal)	RMSE (kcal)	Category Acc (%)
Random Forest	100.00%	**14.27%**	19.57	34.20	90.8%
Linear Regression	70.78%	**72.21%**	59.16	79.00	68.2%
XGBoost	99.67%	**99.63%**	14.63	26.75	92.7%
SVM (SVR)	19.71%	**12.50%**	15.32	31.34	94.0%

Our project pipeline:



Technology & Techniques Used:

- **Language:** Python
- **Libraries:**
  OpenCV, MediaPipe, NumPy, Pandas
  Scikit-learn (Random Forest, Linear Regression, SVM)
  XGBoost, Matplotlib, Joblib
- **Techniques:**
  Pose Estimation, Angle Calculation
  Real-Time Motion Tracking, Rep Counting
- **Models:**
  Linear Regression, Random Forest, XGBoostSVM

- **Modules:**
  Rep Counter + Calorie Burn Predictor

Future Scope:

- The system can be extended to include **more exercises** such as push-ups, lunges, and jumping jacks for a complete fitness solution
- A **user tracking feature** can be added to store daily workout data like reps and calories burned
- The project can provide **progress analysis over time** to help users understand their improvement
- **Graph-based visualization** can be implemented to display workout history, calories burned, and performance trends
- The system can be enhanced with **personalized recommendations and goal setting** for better user experience

References:

[1] D. Bubnis, “Calculating how many calories are burned in a day,” Medical News Today, 01 January 2020. [Online]. Available: https:// www.medicalnewstoday.com/articles/319731.

[2] Models-performance-230449055f15. [15] T. Z. Phyu and N. N. Oo, “Performance Comparison of Feature Selection Methods,” in MATEC Web of Conferences, EDP Sciences, 2016.

[3] “calories_burnt _data,” Kaggle, 2022. [Online]. Available: https://www.kaggle.com/ datasets/sandeepgauti/calories-burnt-data.